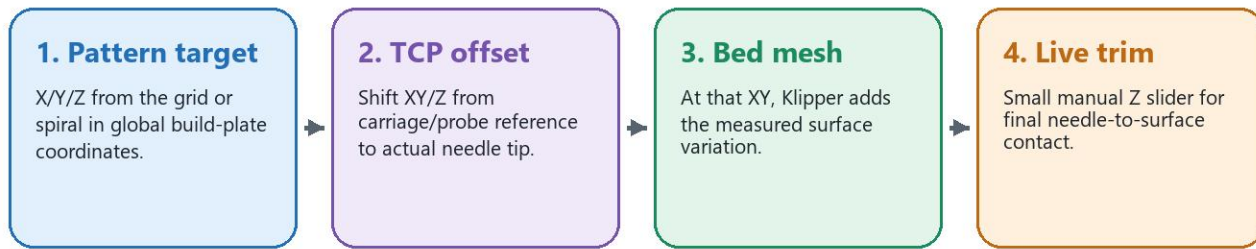


Needle Tip + Bed Mesh Offset Stack

How global pattern coordinates, TCP calibration, bed mesh, and a live manual Z trim combine without rewriting the mesh.



Conceptual physical target

$$X_{\text{cmd}} = X_{\text{pattern}} + \text{tcp_dx}$$

$$Y_{\text{cmd}} = Y_{\text{pattern}} + \text{tcp_dy}$$

$$Z_{\text{cmd}} = Z_{\text{distance_to_surface}} + \text{tcp_dz} + \text{manual_needle_z} + \text{mesh_z}(X_{\text{pattern}}, Y_{\text{pattern}})$$

If the move goes through normal Klipper G0/G1 with bed mesh active, Klipper supplies the mesh_z part internally. The generator or wrapper should only add TCP and manual needle offsets.

Do not rewrite the bed mesh

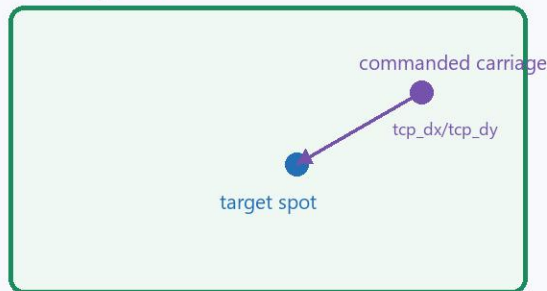
The bed mesh remains the measured surface map. The live needle offset is a separate additive movement correction that can change during a print.

Coordinate Stack at One Spot

The target is the liquid spot coordinate; the machine motion is adjusted so the needle tip arrives there.

XY: target spot vs. carriage command

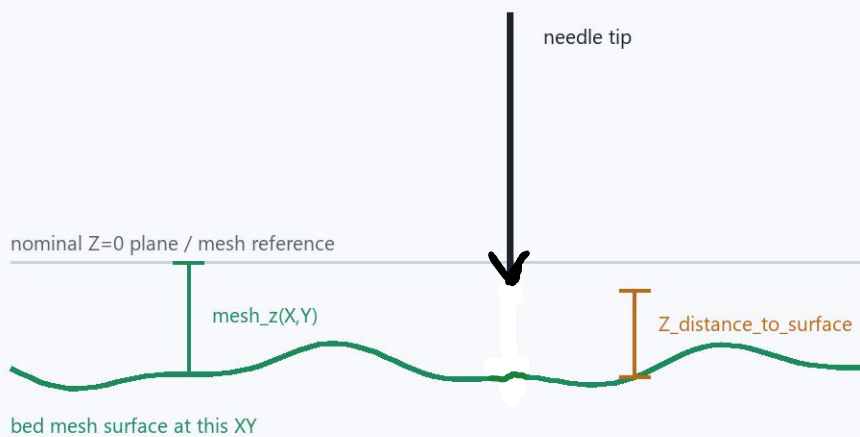
meshed square area



XY rule

The pattern owns the desired liquid position. TCP calibration shifts the carriage so the needle tip, not the reference point, lands on that position.

Z: surface-relative height



Z rule

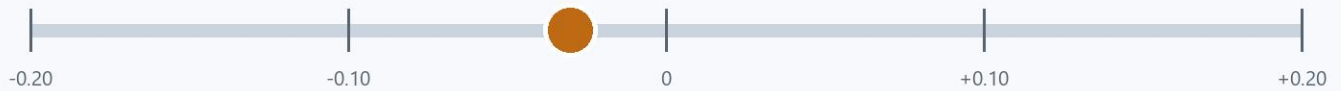
At $Z=0$, the requested needle tip should sit exactly on the local meshed surface. Positive print Z means above surface. The live manual trim nudges this final contact height.

Live Needle Offset Slider

The slider is a fine adjustment layered on top of calibration data, not a new calibration.

Runtime variable

`needle_surface_z_offset = -0.030 mm`



Suggested sign convention: negative = closer to surface; positive = safer / higher.

What changes live

Only the current movement transform/offset changes. With MOVE=1 the machine physically follows the slider while idle or paused.

What stays stable

The bed mesh points and TCP calibration results stay unchanged. The slider can be zeroed without losing calibration.

Operator-facing commands

```
SET_NEEDLE_SURFACE_OFFSET Z=-0.030 MOVE=1 SAVE=1
```

```
SHOW_NEEDLE_SURFACE_OFFSET
```

```
RESET_NEEDLE_SURFACE_OFFSET MOVE=1
```

The GUI slider can send the first command repeatedly. SAVE=1 is optional; use it only when the tuned value should survive restart.

Where to Implement

Decision sheet for keeping the geometry correct and easy to tune.

Layer	Use it for	Risk
Bed mesh	Surface variation only: mesh_z(X,Y)	Do not store needle trim here.
TCP calibration	Measured needle/tool offset: tcp_dx, tcp_dy, tcp_dz	Do not hide manual slider inside calibration results.
G-code move path	Preferred place for live needle Z trim, because bed mesh still applies.	Must use normal G0/G1 path, not raw toolhead.manual_move.
G-code generator	Can add TCP XY/Z to emitted logical moves or emit a named wrapper command.	Less ideal for live tuning during an active print.

Recommended implementation direction

- Keep MESH exactly as surface mapping.
- Keep tcp_offset_x/y/z as calibration data.
- Add a named live variable: needle_surface_z_offset.
- Apply live Z trim through Klipper's normal G-code coordinate path so bed mesh remains active.
- Use a dispensing move wrapper only if it forwards to normal G0/G1 or calls the existing gcode_move transform.

Open decision

Do we want the manual Z trim to affect all Z moves after it is enabled, or only dispensing/contact moves? That decides whether it belongs in SET_GCODE_OFFSET or a dedicated NEEDLE_G0/NEEDLE_G1 wrapper.